EXECUTIVE MEMO

To: CEO / CFO | From: Bethelhem Abay | Date: 02 May 2026 | Re: Preference-Tuned Judge — Week 11 Results

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1 · The Decision

Tenacious's Conversion Engine reached a τ^2-Bench pass_at_1 of 72.67% in Week 10, but five failure probes (A07, E01, E02, E03, G03) revealed repeatable judgment errors; cases where the agent sent outreach it should have suppressed. A preference-tuned ORPO judge was trained on 323 hand-curated pairs across 10 probes and evaluated on 61 sealed held-out examples; it correctly blocked bad outputs 85.2% of the time (95% CI [0.77, 0.93]), with 100% accuracy on the four highest-risk probe categories (A07 disqualifiers, B04 low-confidence funding, D05 soft rejection, G03 escalation).

2 · Headline Lift on Tenacious-Bench Held-Out

Variant	Accuracy	95% CI	Note
A - No judge (baseline)	-	-	All rejected outputs pass through
C - ORPO judge (ours)	85.2%	[0.77, 0.93]	52 / 61 held-out pairs correct
τ^2-Bench baseline	72.67%	[0.65, 0.79]	Week 10 agent, reused per rules

Delta A (ORPO judge vs no-judge): +85.2 pp. The 95% CI [0.77, 0.93] does not include zero, confirming the improvement is not due to chance. Delta B (ORPO judge vs τ^2-Bench baseline) is reported honestly: the held-out evaluation measures catch-rate on known-bad outputs only, not overall agent accuracy; a full τ^2-Bench re-run with the judge in-loop was not completed due to cost constraints.

3 · Cost Per Task

Component	Cost	Notes
Dataset generation (323 pairs)	$0.00	Trace-derived + programmatic, no API
Multi-LLM synthesis (120 pairs)	~$1.50	OpenRouter - DeepSeek, Llama-3
ORPO training (200 steps, T4)	$0.00	Colab free tier, ~17 min
τ^2-Bench baseline (30 tasks × 5)	$0.00	Reused from Week 10
Total (Week 11)	<$1.50	Of $10.00 budget
Cost per training pair	<$0.005	1.50 / 323 pairs
Inference cost per prospect	~$0.00	Local LoRA, no per-call API fee

APPENDIX - The Skeptic's Case

4 · Four Failure Modes Tenacious-Bench v0.1 Does Not Capture

Gap	What is missing	What v0.2 needs
Multi-turn thread coherence	All pairs are single-turn. The agent can fail across a thread (e.g. correct reply 1, wrong reply 2) in ways the judge never sees.	Thread-level pairs spanning 2–4 turns; thread_id as a sequence, not a scalar.
Timezone and scheduling failures	Probes requiring valid send-window validation (07:00–18:00 recipient local time) were declared NOT BUILT in Week 10 and excluded from the dataset. The judge has no signal on timing correctness.	Add a send_timestamp field to context; author 20+ pairs where the rejected output sends outside valid window.
Regulated-industry caveats	Prospects in finance, healthcare, or defence require specific legal hedges. The dataset has no regulated-industry examples; the judge will PASS outputs that are legally non-compliant.	Add an industry_flags field (e.g. regulated:true); author 15+ pairs per regulated vertical.
Implicit opt-out signals	Current opt-out pairs use explicit channel flags. Real opt-outs are often implicit (prospect unsubscribed from a list, email bounced, LinkedIn InMail declined). These are not in the dataset.	Ingest bounce and unsubscribe event logs as signals; author pairs where the disqualifier is inferred, not explicit.

5 · Public-Signal Lossiness in Ground Truth

The dataset's ground truth is only as good as the signals in the context object. Anti-offshore and opt-out flags are sourced from LinkedIn posts, founder interviews, and CRM notes all of which have a signal lag of days to weeks. A prospect who changed their stance after the signal was recorded will be incorrectly suppressed (false positive) or incorrectly sent to (false negative). The held-out evaluation measures judge consistency with the recorded signal, not with the prospect's actual current preference.

6 · One Honest Unresolved Failure from Training

Probe C02 (bench commitment not honoured) achieved only 60% judge accuracy after 200 training steps. The rejected outputs for C02 are syntactically identical to PASS outputs both are 'send' actions with professional tone and the distinguishing signal (a previous commitment in the thread) is buried in the rationale field, not surfaced in the context object. The judge cannot detect this failure because the context object does not contain a structured commitment_made field; the model is being asked to infer a commitment from a prose rationale, which it cannot reliably do with 169 training pairs.

7 · Kill-Switch Trigger Condition

Remove the trained judge component from the send pipeline and revert to the deterministic rule-layer (scoring_evaluator.py, GPT-4o-mini) if any of the following conditions are observed in production:

- False-positive rate exceeds 15% over a rolling 500-prospect window (judge suppressing outputs that a human reviewer rates as acceptable) indicating the model has over-generalised its suppression behaviour.
- Any probe category drops below 50% accuracy on a weekly spot-check of 20 sampled judge decisions indicating distribution shift in prospect signals.
- The adapter fails to load or produces malformed JSON on more than 1% of inference calls indicating infrastructure or quantisation degradation.
- A Tier-1 brand-damage event occurs that the judge did not block, traced to a failure mode present in Tenacious-Bench v0.1 (i.e. not a new failure type) indicating the model has regressed on a known probe.